

# TerraState forecast closure

→ inference

→ training only

→ post-training

## History-only state inference

past meteorology  
+ static  $g$

cloud-masked  
EO history

history-only  $q_\theta$   
context encoder

$P_\rho$

$z_t$

## Weather-driven state transition

full24 future  
weather

static  $g$

horizon  $h$

shared  $T_\psi$

$z_{t+h}$

closure  
 $O_\omega$

$r_h$

+

forecast  
 $\hat{y}_{t+h}$

context-only prior  $b_h$

**Training only:**  $\mathcal{L} = \mathcal{L}_{\text{GT}} + 0.5\mathcal{L}_{\text{KD}} + \lambda_s\mathcal{L}_{\text{future-state}}$

**Forecast target**  
 $\mathcal{L}_{\text{GT}}(\hat{y}_{t+h}, y_{t+h})$

**Frozen full-weather teacher**  
 $\mathcal{L}_{\text{KD}}(\hat{y}_{t+h}, \hat{y}_{t+h}^{\text{teach}})$

**Future-state anchor ( $h = 20$  only)**  
observed future EO →  
frozen  $q_\theta + P_\rho$  / cached  $z_{t+20}^\star$

Same checkpoint  
post-training tests

**Q1 Forecast**  
frozen checkpoint

**Q2 Load-bearing**  
full cut  $T \rightarrow I$

**Q3 Driver response**  
matched mean donor

**Q4 Optional**  
direct · composed · guards